
Privacy-Preserving Machine Learning

An Automated, Data-Driven Approach to Parameter
Selection for Homomorphic Encryption

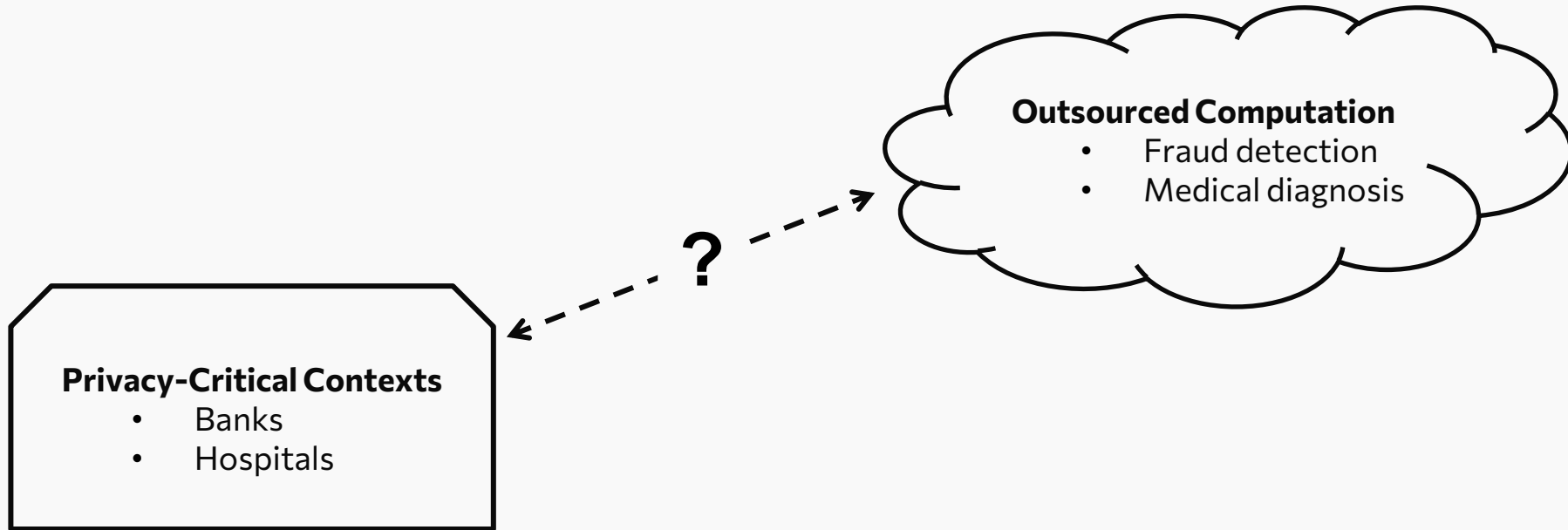
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THE COLLEGE OF
WOOSTER

Motivation

Outsourced computation in privacy-critical contexts



Homomorphic Encryption

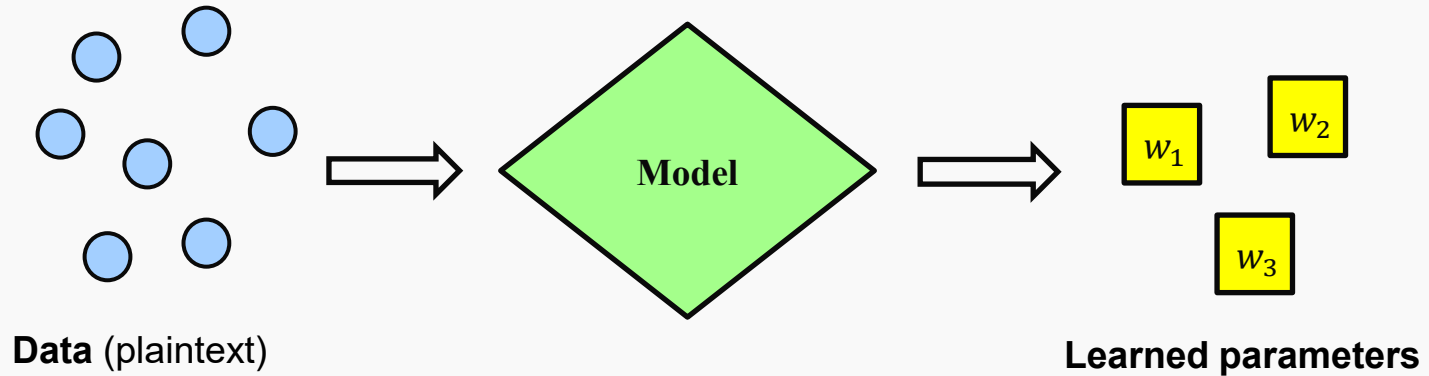
A promising solution

- A cryptographic technique that enables computation directly on encrypted data.
- Under a public-key homomorphic encryption scheme:

$$\text{Dec}_{sk} \left(f \left(\text{Enc}_{pk}(m) \right) \right) = f(m)$$

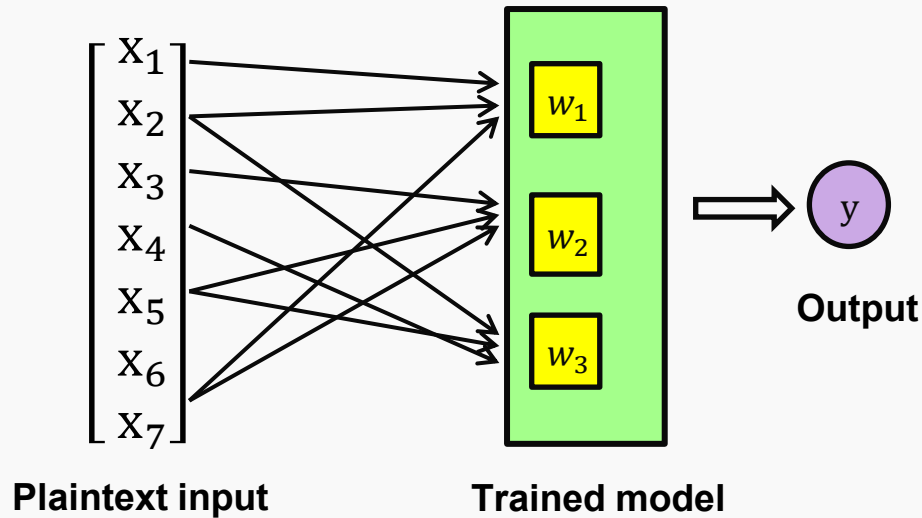
- The supported computations are restricted to addition and multiplication.

Model Training



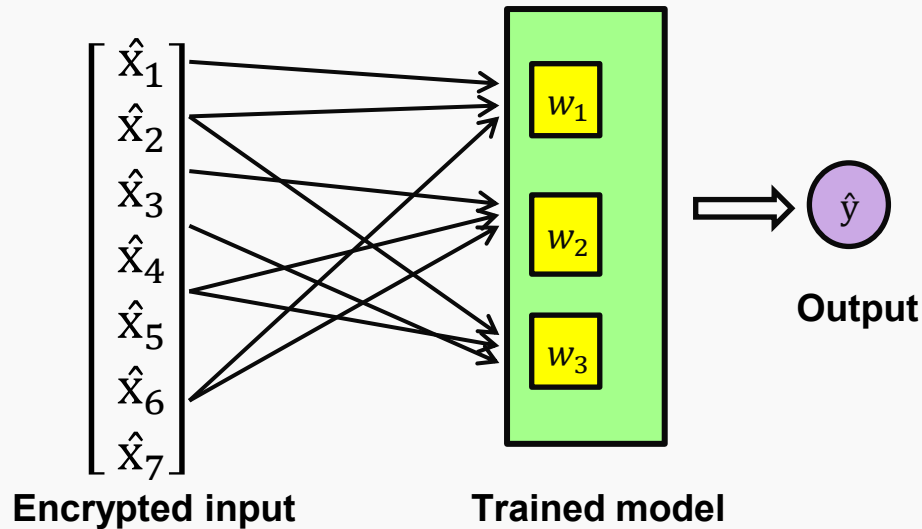
Standard Inference

$$y = x_1w_1 + x_2w_2 + x_3w_3$$



Encrypted Inference

$$\hat{y} = \hat{x}_1 w_1 + \hat{x}_2 w_2 + \hat{x}_3 w_3$$



The CKKS Scheme

- Approximate arithmetic on real-valued data:

$$\text{Dec}(\text{Enc}(x) + \text{Enc}(y)) \approx x + y$$

- Quantum-resistant
- Introduces small noise to be added to ciphertexts
- Real-valued vectors encoded as polynomials

CKKS Parameters

- Three parameters:
 - Polynomial Modulus Degree N
 - Coefficient Modulus Chain Q
 - Scale Δ
- $$\left. \begin{array}{l} \text{Polynomial Modulus Degree } N \\ \text{Coefficient Modulus Chain } Q \end{array} \right\} \mathbb{Z}_Q[X]/(X^N + 1)$$
- $$1.4 \rightarrow 1.4 \cdot (\Delta = 4) \rightarrow 5.6 \rightarrow 6 \rightarrow 1.5$$

→ CKKS parameterization is complex, with choices impacting security, precision, and computational cost.

My Project

An Automated, Data-Driven Approach to CKKS Parameter Selection

- Three key components:
 - Incorporates model and dataset constraints into parameter selection
 - Securely evaluates candidate parameter configurations
 - Frames selection as a trade-off between precision, security, and performance

Dataset Constraint: Decimal Precision

12.34	1.002	60.2	9.3
11.67	1.004	72.7	24.6
9.35	1.001	85.1	63.8
23.46	1.030	34.3	12.4

- Concern: if the difference between two real numbers is less than $\frac{1}{\Delta}$, they may round to the same encoded value:
 - If $\Delta = 100$, $1.002 \cdot \Delta = 100.2 \rightarrow 100$
 $1.001 \cdot \Delta = 100.1 \rightarrow 100$
- Number of bits needed to represent the smallest non-zero gap ($0.001 \rightarrow 10$ bits)

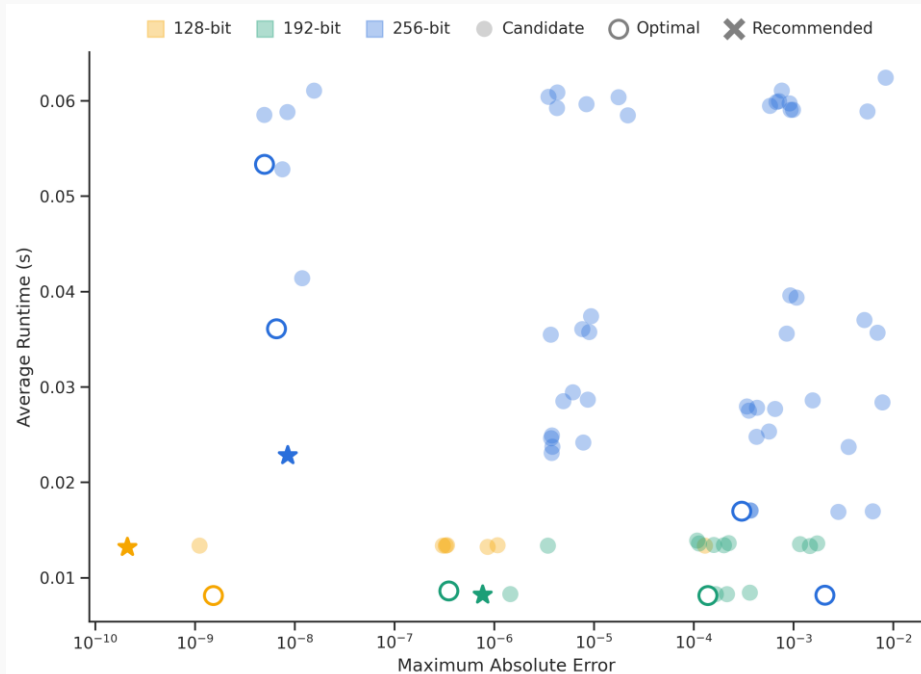
Healthcare Case Study

Parameter Selection as a Trade-off Analysis

- Diabetes dataset
- Simulated real-world encrypted machine learning inference

Healthcare Case Study

Parameter Selection as a Trade-off Analysis



- There is no single optimal solution
- Pareto-optimal configurations

Thank You

